Physical examinations

Physical exam process

- Signalment
- History
- General appearance/ impression (Initial observations)
- Vital signs
- Physical exam (Systems approach)

（http://www.ruralareavet.org/PDF/Physical_Examination.pdf）

Signalment

- Complete description of the animal
- Species, breed, age, sex, reproductive status, other distinguishing characteristics
- Keep this information in mind as you examine the animal and make clinical judgments

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History (Hx)

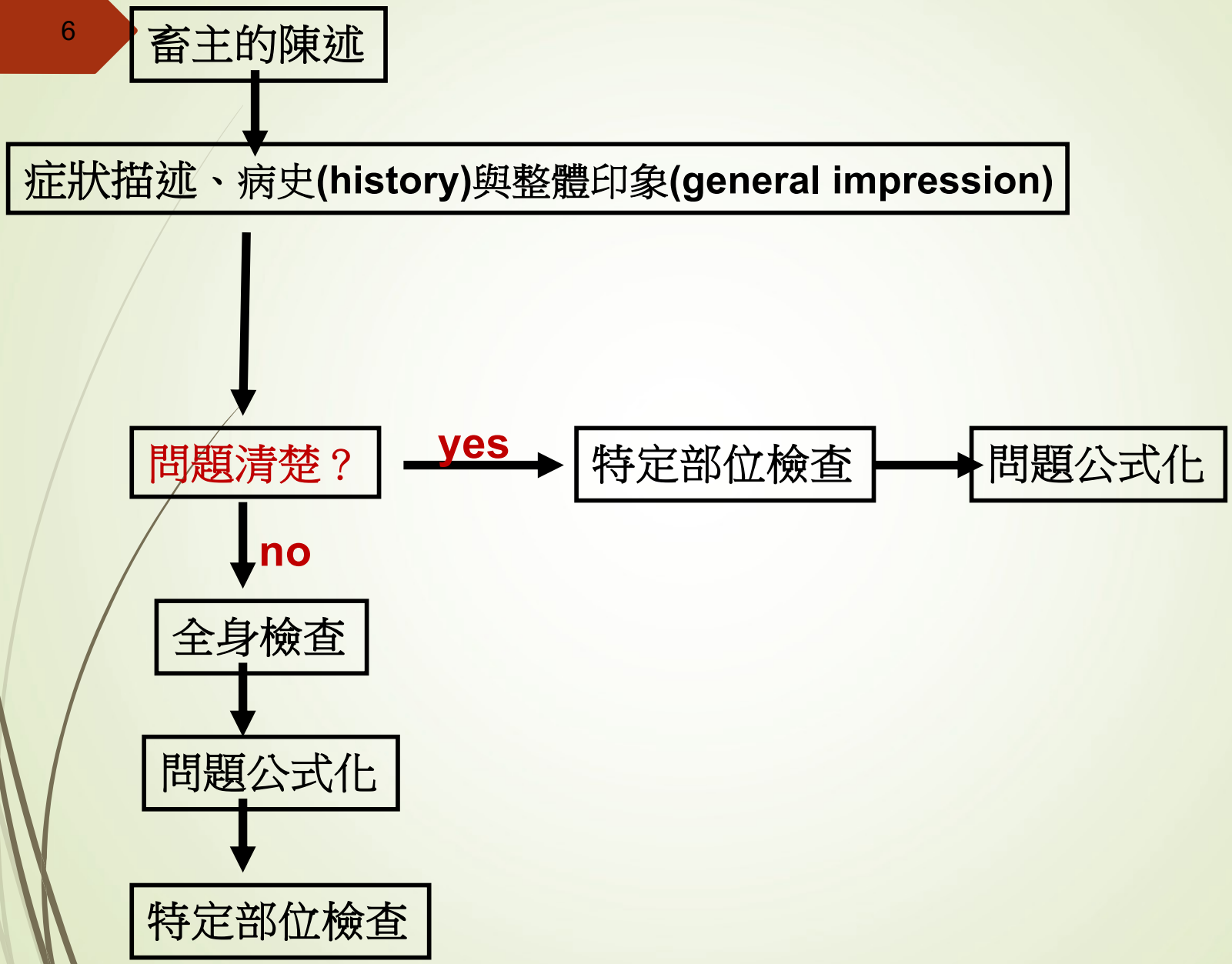
- Includes environment, diet, medical history, reproductive history, vaccination status and current medications
- Chief/presenting complaint
- Reported concerns should be followed up with additional questions to clarify nature of the complaint

（http://www.ruralareavet.org/PDF/Physical_Examination.pdf）

General impression

- Purpose
  - 短時間蒐集適當的資訊
  - 和病史結合
  - 決定診療方向與特定檢查項目

Checking plan



Checking Technique

- The observation and checking are better performed after the visit, talking
- Hair coat will hinder the evaluation of the nutrition level
  - Palpation along the spine

Observation items of general impression 1

- Consciousness
  - Healthy animals should be alert to their surroundings
  - Lethargy, sopor, stupor, coma
- Behavior
  - Most animals should be quiet and cooperative
  - Restless, anxious, aggressive

Observation items of general impression 2

- Posture
  - A seriously ill patient cannot stand
  - Abnormal posture may be seen while standing
  - Sitting, lying, lordosis, kyphosis, neck extended
- Gait
  - Abnormal gaits will be obvious while exercising
  - Stiff, lame, inadequate strength, incoordination

Observation items of general impression 3

- Body form
  - Crossbreeding animals
  - Relation legs/body, Relation abdo./thorax/head
- Nutritional condition
  - Overweight is related to health
  - Obese cat-> lameness and diabetes mellitus
  - Very obese, obese, moderate, poor

Observation items of general impression 3

- coat
  - Poor nutrition and skin disease
  - Dull, thick, thin, alopecia
- Abnormal sounds
  - Short-neck species: breathing sounds
  - Sick animals: loud breathing or GI sounds
  - Joint problems: abnormal sound while walking

Checking procedure form

General Impression

Consciousness	O lethargy	O sopor	O stupor	O coma	O
Behavior	O balanced	O restless	O anxious	O agressive	O
Posture	O sitting O lying	O lordosis	O kyphosis	O neck extended	O
Gait	O stiff	O lame	O inadequate strength	O incoordination	O
Body form	relation legs / body:	O inappropriate	relat. abdom./thorax/head	O inappropriate	O
Nutritional condition	O very obese	O obese	O moderate	O poor	O
Coat	O dull	O thick	O thin	O areas of alopecia	O

Abnormal sounds: .....

Apparent abnormalities:

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Summary of history and general impression → problem formulation

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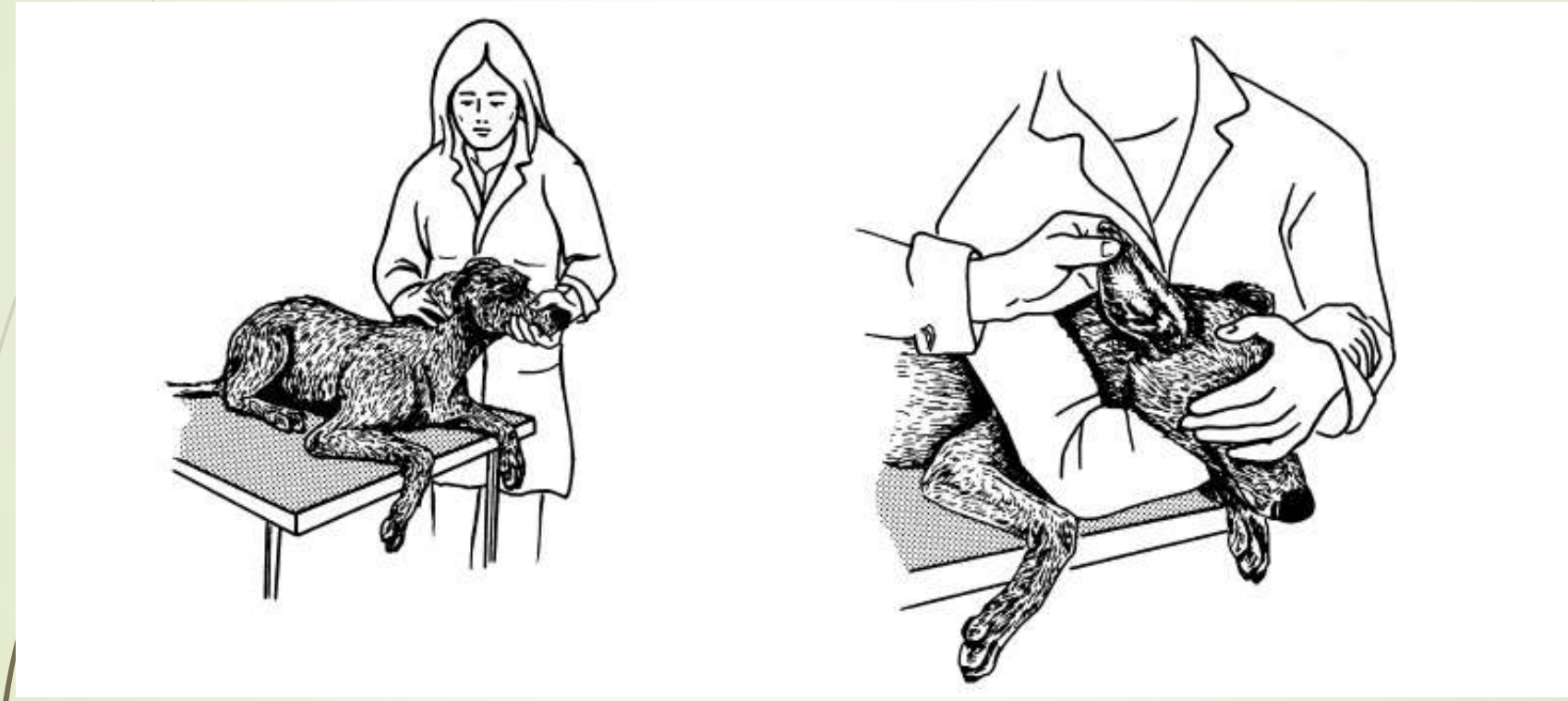
General examination

- Purpose
  - History combined general exam results
    - Formulate the findings
    - Identify the checking direction
    - Detailed specific checking

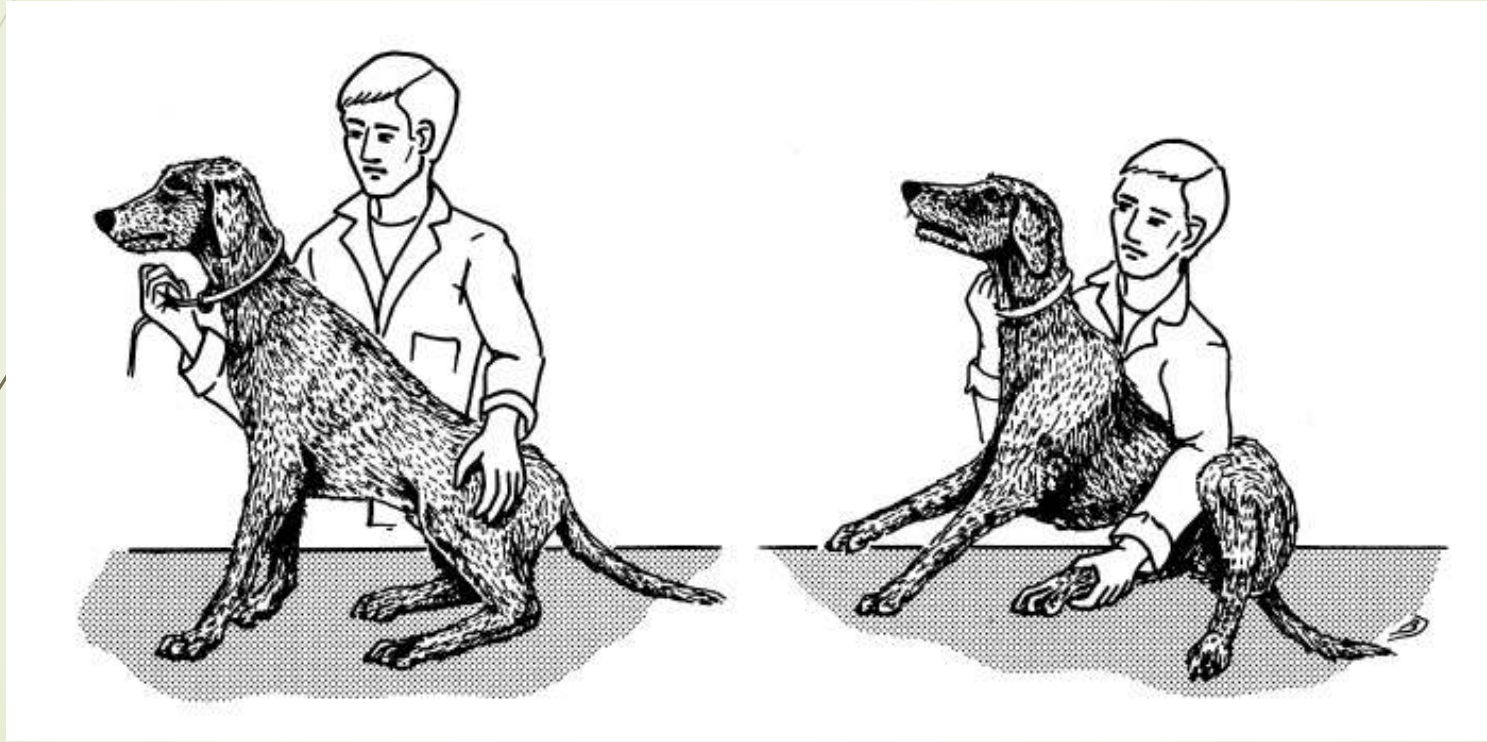
Patient restraint 1

- Help animals relax and let them explore the checking room
- Ask owner to help restrain the animals
- Scratch the hair behind ears of cats to distract the attention

Patient restraint 2



Patient restraint 3



Respiratory movements

- Depth, type, rhythm and frequency
- Normal: costoabdominal respiration
- Abnormal:
  - Pendulous respiration: diaphragm rupture
  - Costal respiration: dyspnea
  - Abdominal respiration: loss of lung elasticity
- Normal frequency

  Dog: 10-30/min

  Cat: 20-40/min

Pulse

- Regularity
  - Regular or irregular
  - Arrhythmia
- Amplitude
  - Magnus（強脈）, parvus （細脈）, filiformis （絲狀脈）, deficit （失脈）
- Form and symmetry
  - Tardos（慢脈）, celer （快脈）, asymmetric frequency
- Normal pulse frequency

Dog: 60-120/min

Cat: 120-180/min

Rectal temperature

- Normal temperature range

  Dog: 38.0-39.0℃

  Cat: 38.5-39.0℃

- BT will elevate 0.5-1.0℃ when animals are excited or nervous

Hair coat

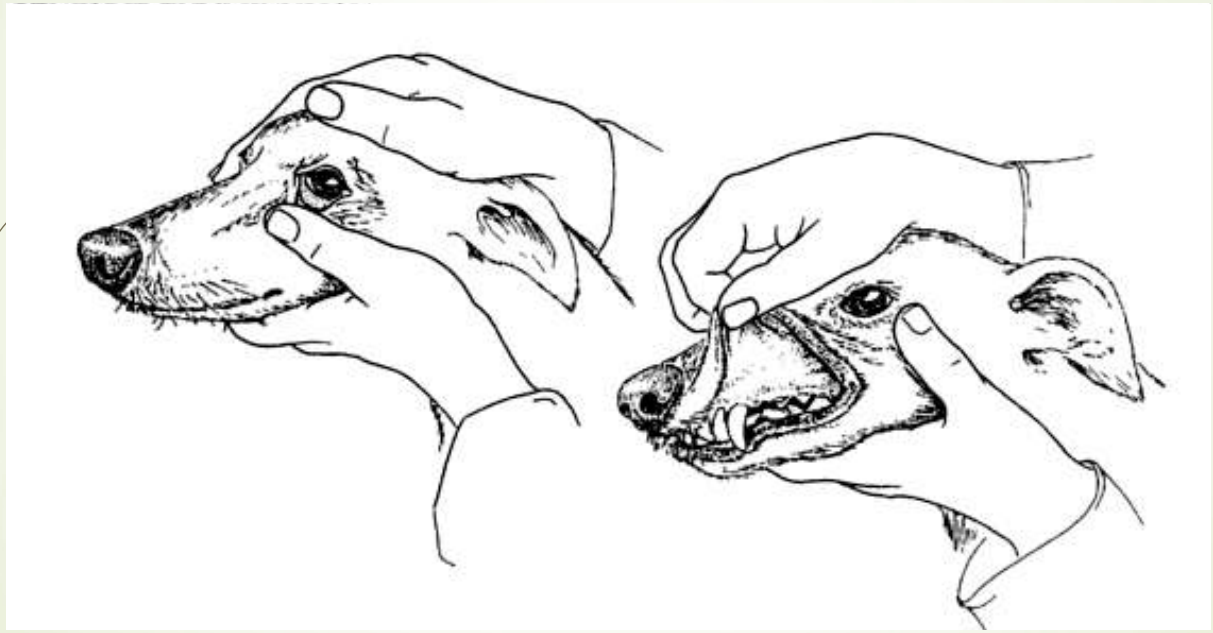
- 毛髮的生長週期：生長期、退化期、休止期
- 生長在一起的毛髮可能處在不同生長階段
- 毛髮的生長受季節與自然光照期的影響
  - 春天較稀疏
  - 家犬全年都在掉毛（人為光照”電燈”的影響）
  - 生病動物蛋白質利用不佳

    ->毛髮稀疏或結構異常
- Coat checking: color change, dull, thin, alopecia, loose hair, ectoparasites

Skin

- Skin color: red, pale, icteric, cyanotic, erythema, bruise, hematoma, pigmented
- Elasticity:
  - 10th肋骨上1/2處
  - Return time: young animal<0.5 sec
    old animal<1sec
    abnormal: >1.5sec
- Skin temperature
- Edema

Mucous membranes

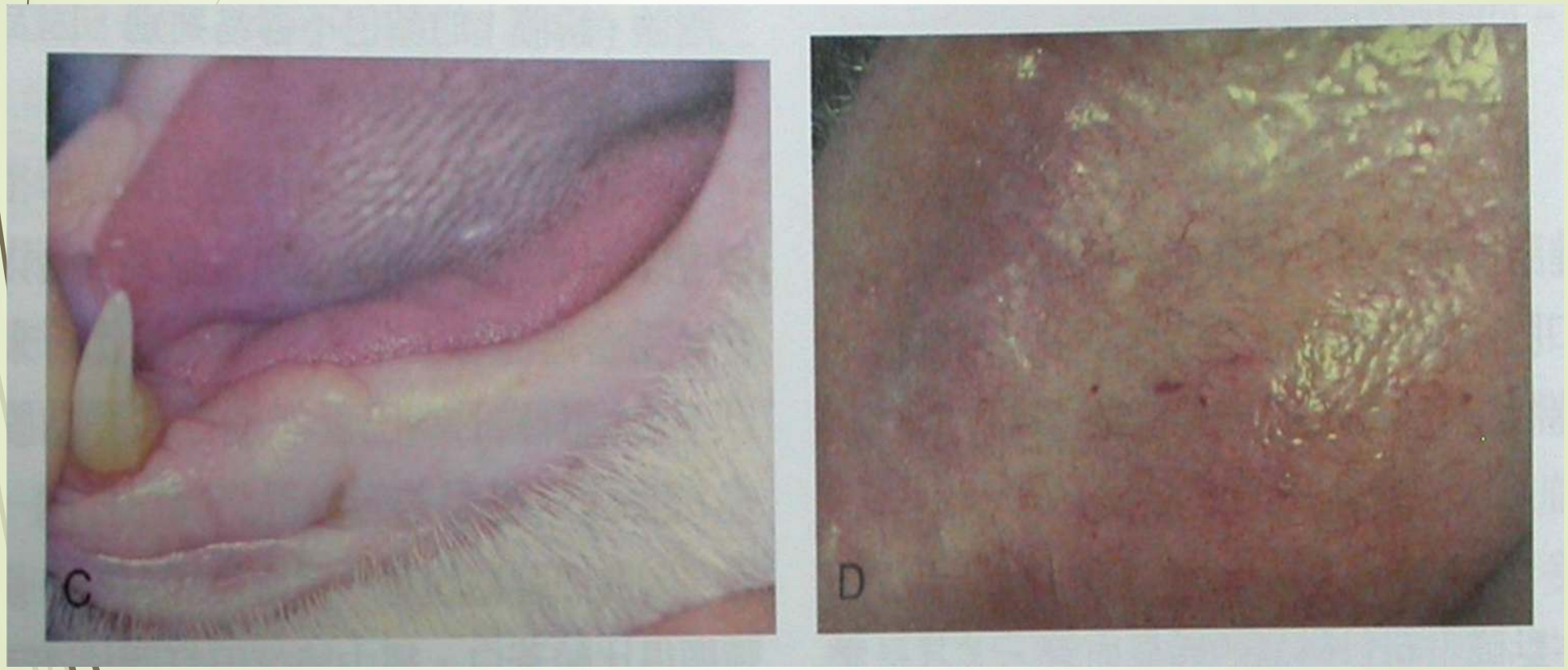


Mucous membranes of dogs





Mucous membranes of cats



Mucous membranes 2

- Mucous membrane color: red, pink, pale, white, icteric, cyanotic
- CRT (capillary refill time): normal< 2 sec
  - CRT> 2 sec:

    compromised circulation due to cold, shock, cardiovascular disease, and anemia
- Hemorrhage or ulcer

Mucous membranes 3

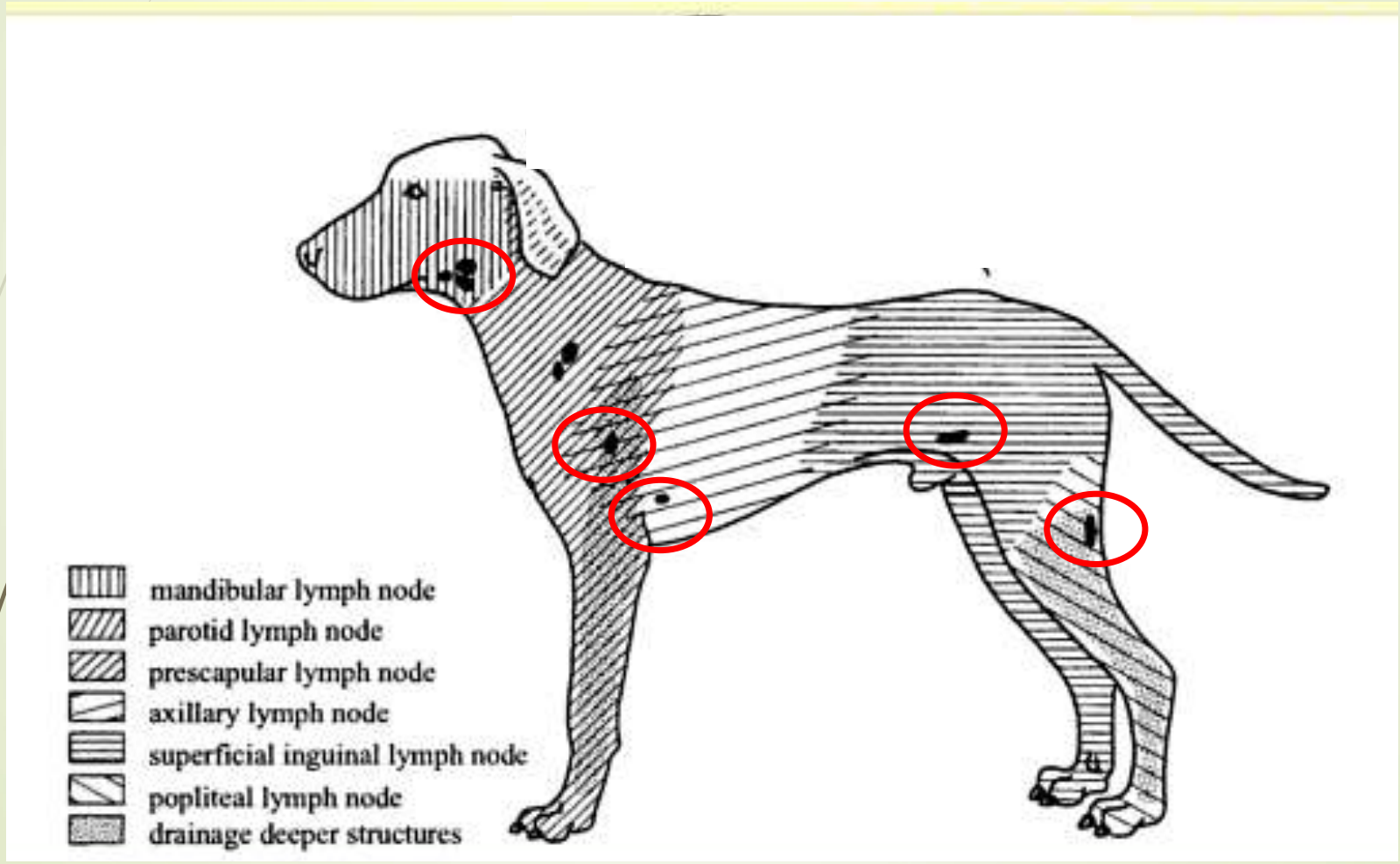
Assessing Mucous Membrane Color

Color	Interpretation	Possible Causes
Pink	Normal	Adequate perfusion/oxygenation of peripheral tissues
Pale or White	Anemia, poor perfusion, vasoconstriction	Blood loss, shock, decreased peripheral blood flow
Blue (cyanotic)	Inadequate oxygenation	Hypoxemia
Brick Red	Increased perfusion, Vasodilation	Early shock, sepsis, fever, systemic inflammatory response syndrome
Yellow (icteric)	Bilirubin accumulation	Hepatic or biliary disorder and/or hemolysis
Brown	Methemoglobinemia	Acetaminophen toxicity in cats, intravascular hemolysis
Petechiae (red splotching)	Coagulation disorder	Platelet disorder, DIC, coagulation factor deficiencies

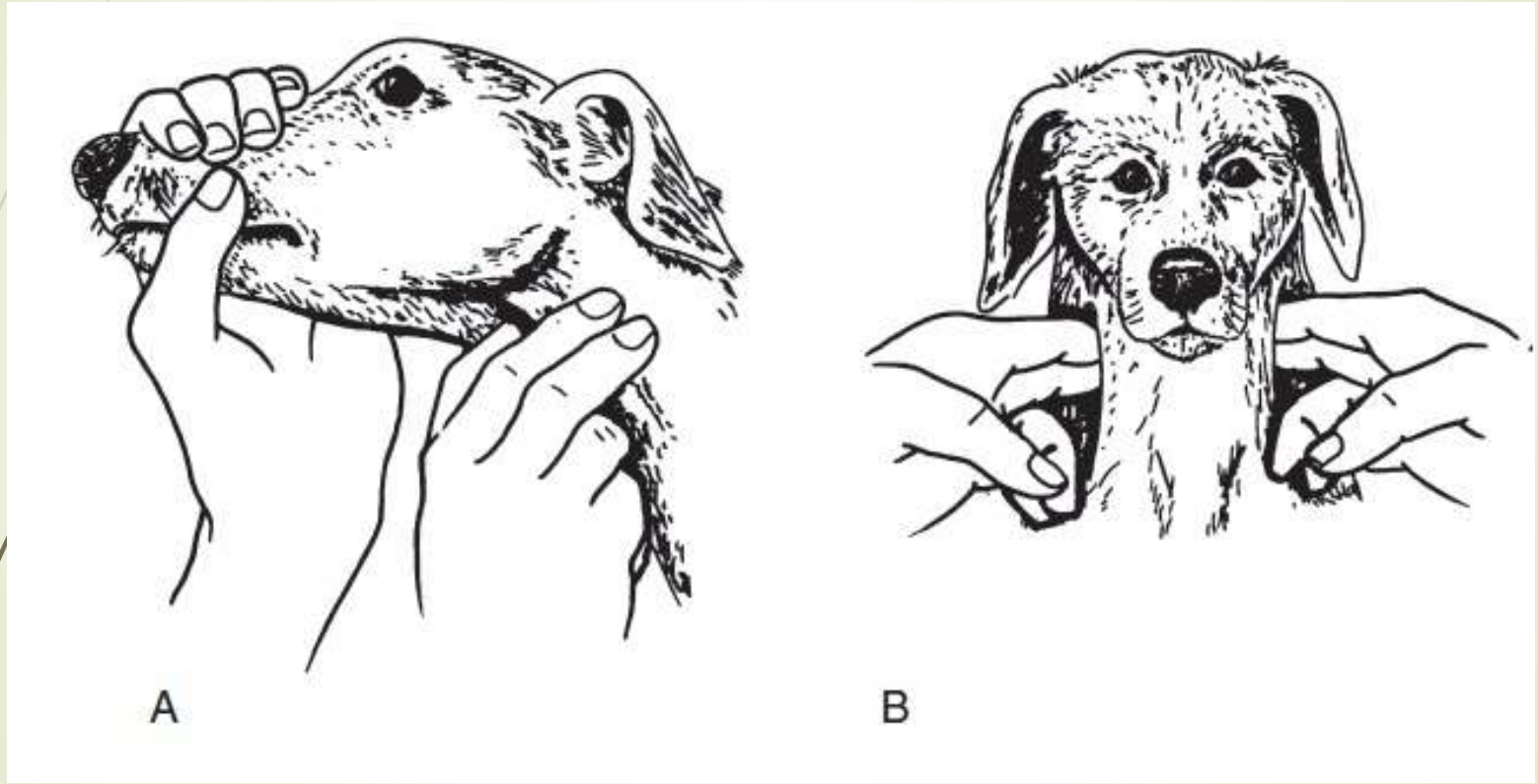
DIC: disseminated intravascular coagulation

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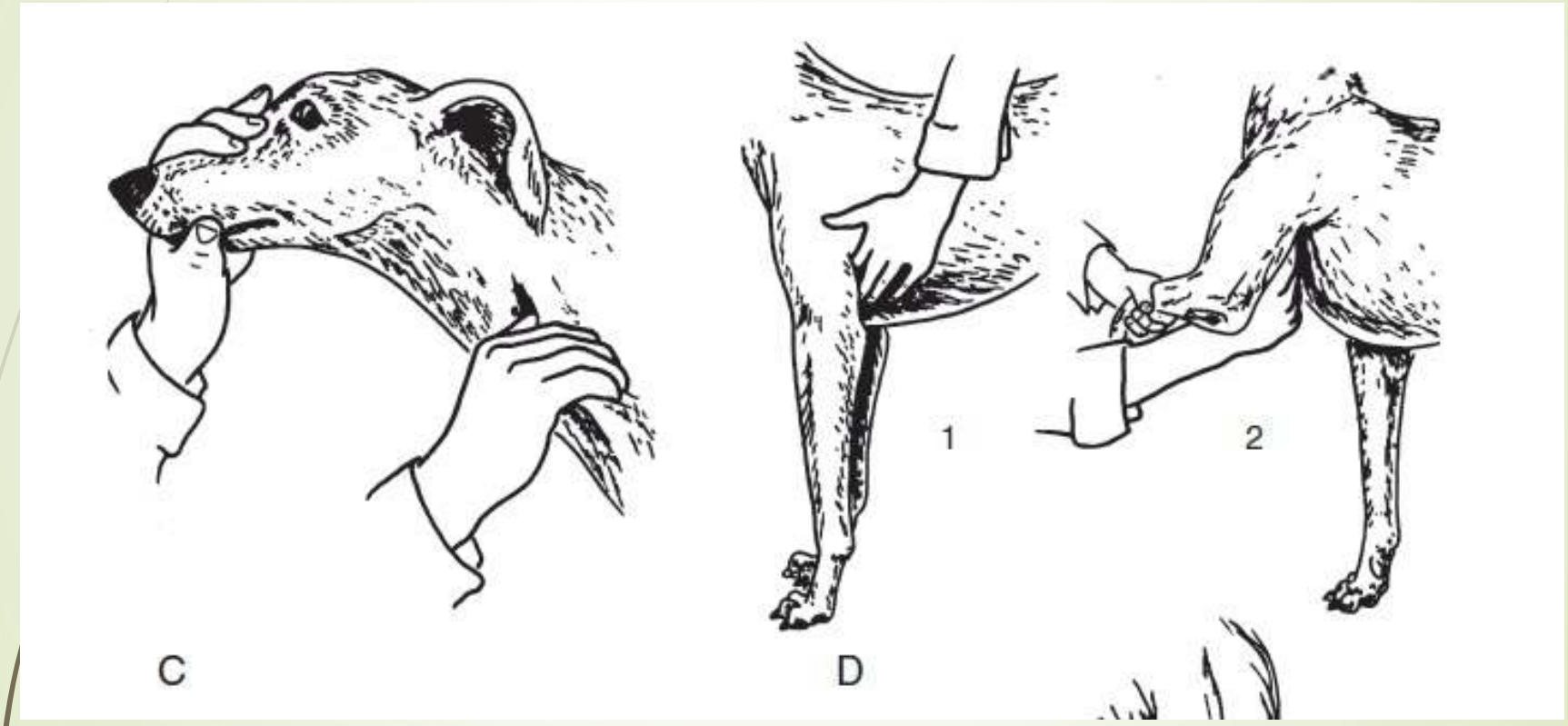
Lymph nodes 1



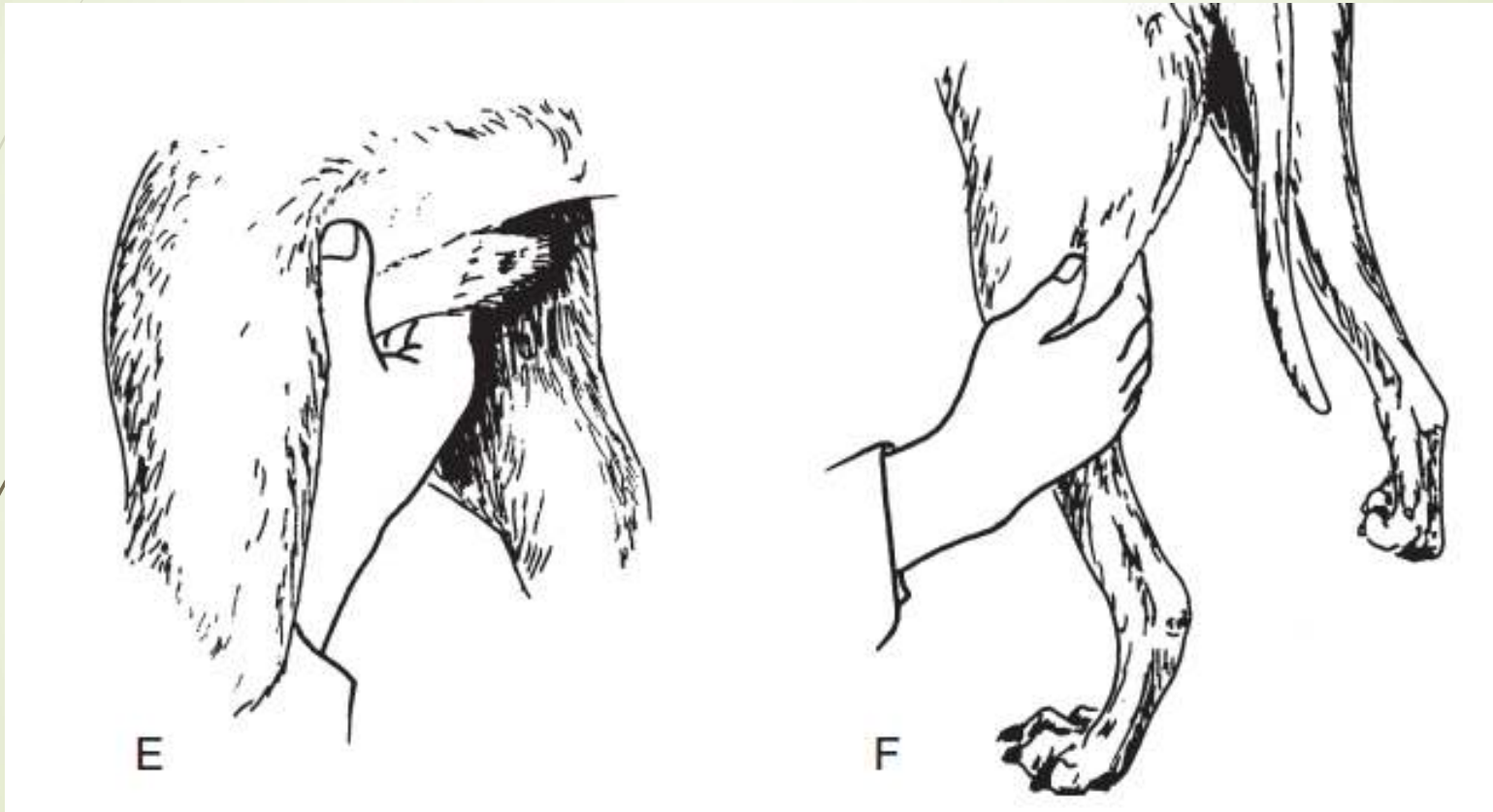
Lymph nodes 2



Lymph nodes 3



Lymph nodes 4



Lymph nodes 4

- Size
- Shape
- Consistency
- Painfulness
- adhesiones

General examination form

General Examination

Respiratory movements					
depth & rhythm	O superfical / panting	O deep	O dyspnea / forced	O irregular	O
type	O costal	O abnormal	O pendulous	frequency / min	O
Pulse					
regularity & rhythm	O inegual	O resp. arrhythmia	O pathol. arrhythmia		O
amplitude & deficit	O p. magnus	O p. parvus	O p. filiformis	O pulse deficit	O
form & symmetry	O p. tardos	O p. celer	O asymmetric frequency / min		O
Rectal temperature ..... °C					
Coat	O color change	O dull O thin	O alopecia O loose hair	O ectoparasites	O
Skin					
color	O red O pale	O icteric	O cyanotic	O pigmented	O
thickness,elasticity	O thin O thick	O inelastic	O return > 1.5 sec		O
temperature,edema	O cool	O warm	O edema (cool)	O edema (warm)	O
Mucous membranes					
color	O red O pink	O pale O white	O icteric	O cyanotic	O
CRT & moistness	O > 1 sec	O dry	O sticky		O
hemorrhage,ulcers	O petechiae	O suffusions	O ulcers		O

Lymph nodes	Mandil. L/R	Retroph. L/R	Prescap. L/R	Axill. L/R	Ing./mam. L/R	Poplit. L/R	
size	.....	.....	.....	.....	.....	.....	O
shape	.....	.....	.....	.....	.....	.....	O
consistency	.....	.....	.....	.....	.....	.....	O
painfulness	.....	.....	.....	.....	.....	.....	O
adhesiones	.....	.....	.....	.....	.....	.....	O

Other notable findings or further descriptions:

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Major references

- Medical History and Physical Examination in Companion Animals, 2nd Edition by A. Rijnberk, F. J. van Sluijs. September 2008. Elsevier.
  - 吳瑞得等翻譯。2010。伴侶動病史與理學檢查。Taiwan Elsevier LLC.
- http://www.ruralareavet.org/PDF/Physical_Examination.pdf

Discussion

- 正常狗貓呼吸、心跳 、體溫的參考值範圍
- CRT>2sec 所代表的臨床意義
- 不同mucous membrane 顏色所代表的臨床意義與可能的原因？